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MILITARY SPECIFICATION

PUMP, CENTRIFUGAL: SELF-PRIMING, DIESEL-ENGINE-DRIVEN, WHEEL MOUNTED, 6-INCH, 1500 GPM AT 60 FOOT HEAD

This specification is approved for use by the USA Belvoir Research and Development Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a diesel-engine-driven close-coupled, self-priming, centrifugal pump mounted on a two-wheel frame.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-P-35	-Passivation Treatments for Corrosion Resisting Steel.
WW-F-406	-Flanges, Cast-Iron (Classes 125 and 250) and Bronze (Classes 150 and 300).

MILITARY

MIL-P-514	-Plates, Identification, Instruction and Marking, Blank.
MIL-T-704	-Treatment and Painting of Materiel.
MIL-D-2940	-Dampeners, Fluid Pressure, Gage Protection.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research and Development Center, ATTN: STRBE-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4320

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MIL-G-3859 - Grease Guns, Hand-Operated, Lever, Push and Screw Type.
MIL-S-8660 - Silicone Compound.
MIL-P-10603 - Pumps and Pumping Units, Centrifugal, Power-Driven, for Water, Packaging of.
MIL-G-10924 - Grease, Automotive and Artillery.
MIL-R-12144 - Reflector, Indicating, Clearance.
MIL-G-23827 - Grease, Aircraft and Instrument, Gear and Actuator Screw.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.
FED-STD-292 - Truck, Light Commercial, Gasoline/Diesel Fueled, Four Wheel Driven (4 X 4, 3,000 to 10,000 pounds GVW).

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130 - Identification Marking of US Military Property.
MIL-STD-209 - Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.
MIL-STD-461 - Electromagnetic Interference Characteristics Requirements for Equipment.
MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-889 - Dissimilar Metals.
MIL-STD-1188 - Commercial Packaging of Supplies and Equipment.
MIL-STD-1410 - Methods for Selection of Industrial Engines for End Item Application.
MIL-STD-1472 - Human Engineering Design Criteria for Military Systems, Equipment and Facilities.
MIL-STD-1474 - Noise Limits for Army Materiel.
MS15003 - Fittings, Lubrication (Hydraulic) Surface Check, 1/8 Inch Pipe Threads, Steel, Type III.
MS33800 - Connector, Receptacle, Electrical - Pin Contact, No. 12, 14 and 16 A.W.G., Waterproof.
MS35000 - Battery, Storage, Lead-Acid, Waterproof.
MS35644 - Filler Neck: Fuel Tank - Military Vehicles.
MS35645 - Cap, Fuel Tank: Tactical Vehicle, Fording.
MS35647 - Padlocks, Key Operated.
MS35829 - Hinge, Continuous (Piano), .075 Thickness, .187 and .250 Pin Dia.

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MS51339 - Coupler, Drawbar, Ring-Offset.
MS52125 - Composite Light - Tail, Stop, Turn and Marker.

DRAWINGS

AT

7070301 - Chart, Electrical, Circuit, Numbers.
8347200 - Schematic, Wiring Diagram, Cart-type, Trailer, 24 Volts (Typical).

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DEPARTMENT OF TRANSPORTATION (DOT)

Motor Carrier Safety Regulations.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

ENVIRONMENTAL PROTECTION AGENCY (EPA)

40 CFR, Part 85 - Control of Air Pollution From New Motor Vehicles And New Motor Vehicle Engines.

(Application for copies should be addressed to the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.)

STATE OF CALIFORNIA AIR RESOURCES BOARD

Health and Safety Code.

(Application for copies should be addressed to the California Air Resources Board, 1709 11th Street, Sacramento, CA 95814.)

(Copies of specifications, standards, and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERCIAN WELDING SOCIETY, INC. (AWS)

D1.1 - Structural Welding Code, Section 5, Qualification.

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(Application for copies should be addressed to the American Welding Society, Inc., 2501 N.W. Seventh Street, Miami, FL 33125.)

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

B40.1 - Gages-Pressure and Vacuum, Indicating Dial Type-Elastic Element.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A36 - Structural Steel.
A48 - Gray Iron Castings.
A276 - Stainless and Heat-Resisting Steel Bars and Shapes.
A582 - Free Machining Stainless and Heat-Resisting Steel Bars, Hot-rolled or Cold-finished.
B584 - Copper Alloy Sand Castings for General Applications.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

HYDRAULIC INSTITUTE

Standards for Centrifugal, Rotary and Reciprocating Pumps.

(Application for copies should be addressed to the Hydraulic Institute 122 East 42nd Street, New York, NY 10017.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

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3. REQUIREMENTS

3.1 Description. The pump unit shall consist of a self-priming centrifugal pump, an intermediate bearing housing, diesel engine and other accessories mounted on a two-wheel frame.

3.2 First article (preproduction model). The contractor shall furnish one or more pumps for examination and testing within the time frame specified (see 6.2), to prove prior to starting production that his production methods and choice of design detail will produce pumps that comply with the requirements of this specification. Examination and tests shall be as specified in section 4; and, unless otherwise specified herein, all examinations and tests shall be conducted by the contractor subject to surveillance and approval by the Government (see 6.3). When specified (see 6.2), the Government will conduct any or all of the first article examinations and tests, as specified (see 6.2).

3.3 Material. Materials shall be as specified herein and on the applicable drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.9).

3.3.1 Material deterioration and control. The pump shall be fabricated from compatible materials inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environment to which the item may be exposed.

3.3.2 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.3 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with components and sub-components, and shall make information available upon request to the contracting officer or designated representative.

3.4 Threaded parts. All screw threads shall be in accordance with FED-STD-H28.

3.5 Environmental conditions.

3.5.1 Operating temperature. The pump unit shall perform as specified herein in any ambient temperature from +120° F to +32° F.

3.5.2 Storage temperature. The pump unit shall not be damaged by storage under the following conditions:

High temperature - 155° F, 4 hours daily.
Low temperature - minus 65° F, 12 hours without benefit of solar radiation.

3.6 Safety. All belts, gears, pulleys, shafts, flywheel and other moving parts shall be guarded when such parts are exposed to contact by personnel or otherwise create a hazard. All hot surfaces, including exhaust pipes and other

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lines which may be subject to high temperatures, exposed to contact by personnel, or which create a fire hazard, shall be fully guarded or insulated. The fuel tank shall be located in a manner which will not allow spills or overflows to run on the engine, exhaust pipes or electrical equipment. All points requiring lubrication during operation shall have fittings located or guarded so as to be accessible without hazardous exposure to personnel performing these operations. Live parts of wiring and electrical equipment shall be guarded to protect all persons and objects from electrical shock hazard. Exhaust or discharge from equipment shall be directed so that they do not endanger personnel.

3.7 Human factors engineering. The pumps shall conform to the applicable human factors engineering criteria as specified in MIL-STD-1472, and be operable by 5th through 95th percentile soldiers who wear clothing appropriate to the environment of interest (see 3.5.1) in daylight darkness.

3.7.1 Noise limits. The noise level at the operator's position of the pump shall be within 85dB(A) when tested as specified in 4.5.2.5. If the noise level exceeds 85dB(A), the distance from the pump at which 85dB(A) is met shall be determined. That distance shall be shown on a caution sign mounted on the pump, stating "CAUTION, hearing protection required within _____ feet when equipment is operating." The caution shall be legible from the distance shown on the sign. The caution sign shall be in accordance with MIL-STD-1474 except as modified above.

3.8 Maintainability.

3.8.1 Maintenance support. All assemblages or support elements such as technical manuals, repair parts, special tools, or lubrication charts, as specified (see 6.7 and 6.8), shall be adequate to perform the intended function of assisting or conducting maintenance operations when tested as specified in 4.5.2.7.

3.8.2 Maintenance ratio. The pump shall have a maintenance ratio of not more than 0.08. Maintenance ratio is defined as the ratio of the total active maintenance man-hours required (scheduled and unscheduled) to the total operating time. Man-hours for repair of replaced components and scheduled before-and-after operation checks are excluded. A maintenance schedule shall be furnished prior to the start of any testing.

3.9 Reliability. The specified mean-time-between-failure (MTBF) shall be 200 hours when the pump is tested as specified in 4.5.2.6.

3.10 Pump. The pump shall be an integrally self-priming (after case has been initially filled) centrifugal pump. The pump shall be provided with an internally NPT-threaded pump-case drain port, and a provision for filling the pump case without removal of suction or discharge connections. The pump suction and discharge ports shall have 6-inch, plain-faced flanges and threaded companion flanges conforming to WW-F-406, class 125. The pump suction direction shall be horizontal. The pump discharge shall be horizontal, which may be accomplished either by the design of the pump case or by the use of suitable

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fitting. The pump shall be provided with a corrosion-resisting, self-lubricated mechanical shaft seal. The pump shall pass spherical solids 1-1/2 inches in diameter. The pump shall be mounted on the rear portion (opposite lunette end) of the two-wheel frame.

3.10.1 Pump case. The pump case shall be provided with mounting feet and shall be fabricated of cast iron conforming to ASTM A48, class No. 30. The pump case shall be provided with a replaceable wear plate fabricated of bronze conforming to ASTM B584, alloy 903.

3.10.2 Impeller shaft. The impeller shaft shall be fabricated of corrosion-resisting steel conforming to ASTM A276, type UNS 541000, condition H or steel conforming to ASTM A582, type 41600, condition H or steel conforming to ANSI type 4140, cold finished with an ultimate tensile strength of not less than 130,000 psi and a Brinell hardness range of 270-320. When the impeller shaft is not made of corrosion-resisting steel, a pump shaft sleeve for mounting the shaft seal shall be provided. The sleeve shall be fabricated of corrosion-resisting steel conforming to ASTM A276, type 304. All corrosion resisting steel shall be passivated as per QQ-P-35.

3.10.3 Impeller. The impeller shall be semiopen vane nonclogging type fabricated of bronze conforming to ASTM B584, alloy 836. The impeller shall be keyed to the impeller shaft and held in position by a lock-cap nut with a shaft-protecting washer, or shall be screwed on the shaft in a direction opposite to the direction of rotation. The impeller shall be balanced to within 0.25 ounce-inch (oz-in) at the normal operating speed of the pump.

3.10.4 Intermediate bearing housing. Close-coupling shall be accomplished through the use of an intermediate bearing housing. The housing shall mount directly to the engine flywheel housing and to the pump case. The housing shall be provided with antifriction bearings of such size and type so as to absorb the maximum radial and thrust loads imposed by the pump impeller. The bearings shall be either sealed for life or in a grease-filled enclosure that will not require periodic refilling except when the bearings are removed from the enclosure. The housing shall be fabricated of cast iron conforming to ASTM A48, class No. 30.

3.10.5 Coupling. A flexible- or spline-type coupling shall be provided which shall enable the pump and bearing housing to be removed from the engine as a unit by uncoupling the drive half from the driven half of the coupling. The coupling shall be of the dry-type and shall provide for misalinement and end float. The coupling shall transmit power from the engine to the impeller shaft through an arrangement which will reduce transmission of torsional vibration. The coupling shall have a horsepower rating in excess of the maximum brake horsepower required by the pump.

3.10.6 Check valve. A check valve shall be provided on the suction inlet of the pump. The check valve shall be located on the pump-case side of the suction flange. When subjected to a static discharge head equal to the maximum total head indicated on figure 1, the valve shall prevent siphoning of liquid from the

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pump case and shall not leak or show permanent deformation of the body, cover, seat, or any of its components.

3.10.7 Hydrostatic pressure. The pump shall not leak or show permanent deformation when subjected to an internal hydrostatic pressure equal to 1-1/2 times the shutoff head developed at full throttle operation.

3.10.8 Performance characteristics. When operating at conditions equivalent to standard sea level atmospheric conditions of 29.92 inches of mercury using water with a specific weight of 62.3 pounds per cubic foot at a temperature of 68° F, the pump shall perform as specified in 3.10.8.1 through 3.10.8.3.

3.10.8.1 Priming.

a. Prime in not more than 3 minutes when subjected to a static suction lift equivalent to 25 feet using 30 feet of 6-inch suction line and one elbow.
b. Prime in not more than 1 minute when subjected to a static suction lift equivalent to 10 feet using 15 feet of 6-inch suction line and one elbow.

3.10.8.2 Head capacity. The pump shall deliver not less than the capacities shown in figure 1 against a static suction lift equivalent to 10 feet when running at an operating speed not to exceed 2600 revolutions per minute (rpm). In addition, the pump shall deliver not less than 750 gallons per minutes (gpm) against a minimum total head capacity using a static suction equivalent to 25 feet.

3.10.8.3 Pump efficiency. The pump shall develop a peak efficiency of not less than 65 percent while meeting the requirements of figure 1.

3.11 Wheel mounting. The pump unit shall have a gross vehicle weight of not more than 3500 pounds. The lunette load shall be not more than 200 pounds nor less than 75 pounds. The pump unit shall conform to the following dimensions:

Overall length, maximum	120 inches
Overall width, maximum	75 inches
Overall height, maximum	70 inches
Ground clearance, minimum	12 inches
Tread, axle, minimum	62 inches

3.11.1 Mobility characteristics. After being towed by a 3/4-ton, 4-by-4, truck (see 3.22), the pump unit shall perform without malfunction, damage, or permanent deformation to any of its components under the following conditions (see 6.6):

a. Cross-country operation - Over open, rolling, hilly or cross-country terrain at speeds up to 20 mph.
b. Unimproved road operation - Over unimproved and secondary roads at speeds up to 30 mph.
c. Highway operation - Over smooth, hard-surfaced roads at speeds up to 50 mph.

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3.11.2 Trailing and turning ability. When coupled to a 3/4-ton, 4-by-4, truck (see 3.22), the pump unit shall track the truck without bottoming, weaving, or sideswaying. The turning clearance of the pump unit shall be such that when the pump unit becomes jackknifed, its longitudinal centerline shall form a minimum horizontal angle of 60 degrees to the right and 60 degrees to the left with the longitudinal centerline of the truck. Turns shall be performed without interference or damage to either the pump unit or the truck at the truck's minimum turning radius.

3.11.3 Frame. The frame shall be fabricated of structural steel conforming to ASTM A36 and shall be suitable for the installation of the pump, engine, and running gear.

3.11.4 Lunette and support legs. The pump unit shall be equipped with a lunette and three support legs. The support legs shall have sufficient strength and rigidity to support the pump during operation with the wheels clear of the ground.

3.11.4.1 Lunette. The lunette shall conform to MS51339-1 or -2 and shall be installed so that the offset of the eye may be reversed.

3.11.4.2 Front support leg (lunette end). A front support leg and the associated mounting bracket shall be provided and shall be installed such that the level ground height of the lunette rotated to the lower position is 24 inches, plus or minus 1/2 inch. The front support leg shall fold to the travel position and shall not reduce the angle of approach when folded. Any locking pins used to secure the leg in the up or down position shall be secured by a captive chain or other means to prevent their loss. The ground-bearing surface of the front support leg shall contain not less than 25 square inches and shall be flat.

3.11.4.3 Rear support legs. Two rear support legs shall be provided to prevent the frame from tipping over when hose or pipe is coupled to the pump or when pump or bearing housing is being filled or serviced by personnel. The legs shall be capable of being securely locked in both the retracted and extended positions. The locking devices and the legs shall be captive-chained to the pump unit frame to prevent loss. The legs shall be capable of being adjusted in 2-inch increments from the fully retracted position to the fully extended position, and vice versa, by hand. When in the fully retracted position, the jack stands shall not protrude more than 2 inches below the bottom of the side members of the frame. The ground-bearing surfaces of the rear support legs shall contain no less than 16 square inches per leg.

3.11.5 Intervehicular connectors. Intervehicular connectors shall be installed on the pump unit so that they shall be extended between 36 and 42 inches toward the prime mover from the center of the lunette eye. A clip shall be furnished to hold the electrical harness plug and to double the harness back on the tongue or frame.

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3.11.6 Safety chains. The pump unit shall be equipped with two safety chains. The chains shall conform to SAE J684, class 4, except that the length shall be the same as that for the intervehicular connectors.

3.11.7 Axle assembly. The axle assembly shall include spindle, hub, and wheel components. Wheel brake-assembly components are not required. The load rating of the axle assembly, after being derated to compensate for cross-country operation, shall be no less than the gross vehicle weight.

3.11.8 Suspension. The axle suspension system shall be of a springing capacity to sustain the rated payload under the performance characteristics specified herein without transmitting damaging shock to the pump unit. The suspension shall result in as low a center of gravity as possible without affecting the ground clearance requirements. Rubber bumper stops shall be provided to prevent metal-to-metal bottoming of the suspension.

3.11.9 Shock absorbers. Two shock absorbers shall be furnished and shall control jounce and rebound. The shock absorbers shall be mounted on the frame with control linkage connected to the axle at each side of the pump unit.

3.11.10 Wheels, tires, and tubes. The wheels, tires, and tubes shall be of sufficient size to maintain the minimum ground clearance specified herein. Load ratings for each tire shall be not less than 1/2 the axle rating before the axle rating is derated for cross-country operation. Sizes of tires, wheels, and tubes shall be selected to provide the proper load-carrying capability without exceeding a 50 psi tire air pressure. Tires may be tubeless type without tubes and the tread is optional. The tire inflation pressure shall be labeled adjacent and above the tires.

3.11.11 Wheel-assembly balancing. Each assembly of tire, tube, wheel, and hub components shall be balanced to preclude wheel tramp and drumming at road speeds up to 50 mph.

3.11.12 Chock blocks and cradles. The pump unit shall be furnished with two chock blocks and two cradles. The chock blocks shall be captive-chained to the pumping unit with chains of sufficient length to permit placement of the chock blocks in front of or behind the pump unit wheels. The chock blocks shall conform to SAE J348. The cradles shall secure the chocks to prevent independent movement of the chocks while stowed.

3.11.13 Electrical systems. The pump unit shall be equipped with two separate electrical systems. One shall be energized by the towing vehicle and shall include lights associated with vehicular transportation; the operating electrical system shall include batteries for engine starting and operation plus lights for the control panel and power for electrical devices. The operating electrical system shall be as specified in 3.13 and 3.14.

3.11.13.1 Vehicular wiring and connections. The vehicular electrical system shall be energized through the intervehicular connector (see 3.11.5). Wherever possible, wiring shall be bundled, inserted in a loom, or taped continously and

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secured to the pump unit at not more than 18-inch intervals. The wiring shall be routed in protected spaces away from excessive heat or accidental damage. Connectors shall conform to MS33800 and shall be lubricated prior to assembly with paste-type electrical insulation compound conforming to MIL-S-8660. Grommets shall be installed where wiring passes through sheet metal and frame members. The circuitry shall conform to the wiring diagram on drawing 8347200. Conductors shall be numbered in accordance with drawing 7070301.

3.11.13.2 Vehicular lighting. The pump unit shall be equipped with a service and blackout lighting system. The service lighting shall conform to the requirements of the DOT Motor Carrier Safety Regulations. Two taillamps conforming to MS52125 shall be installed at the rear of the pump unit as far apart as possible to indicate vehicle width and for turn-signal operation. A blackout stoplight conforming to MS52125 shall be installed to the left of the right taillamp.

3.11.14 Reflectors. Reflectors conforming to MIL-R-12144, types I and II, shall be furnished and shall be mounted in accordance with DOT Motor Carrier Safety Regulations. In addition, one other reflector shall be mounted facing forward on each front corner; and one red reflector shall be mounted on each side near the rear of the pump unit and shall be installed at approximately the same height as those installed on the rear. All reflector-mounting locations shall give the maximum reflected warning and shall be afforded protection from obstacles and damaging shock.

3.12 Toolbox. A toolbox made of steel not less than 0.071 inch in thickness shall be attached to the pumping unit. The toolbox shall be of sufficient size to contain the technical manuals furnished with the pump and operator maintenance tools consisting of a 12 inch adjustable open end wrench, a 6 inch shank by 5/16 inch blade screwdriver, a one pound ball peen hammer, a pair of 6 inch slip joint pliers and any special operator maintenance tools. Only special operator maintenance tools shall be furnished and placed in the tool box. The toolbox lid shall be secured to one edge of a side by a continuous hinge conforming to MS35829-7C extending the full length of the box edge. The lid shall be held in the closed position by not less than two fasteners. The hasp shall be located to secure the lid to the box and accommodate a lock conforming to MS35647. The toolbox shall be mounted in an accessible, protected location permitting the lid to be opened through an arc of 90 degrees. The lid shall be held fully open by the weight of the lid or by a fastener. Drainage of the toolbox shall be provided.

3.13 Diesel engine. For the purpose of selecting the engine, the pump unit shall be classified as a class II end item in accordance with MIL-STD-1410. The engine, including all systems, components and accessories, shall conform to MIL-STD-1410, unless otherwise specified herein. The engine shall be air cooled. The maximum governed speed for each engine, as applied, shall not exceed the engine manufacturer's recommended speed for this application. All engine accessories shall be furnished by or approved by the engine manufacturer for this application. The engine shall be capable of operating on fuels and lubricating oils as specified in MIL-STD-1410. The engine shall comply with the

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Environmental Protection Agency and the California Air Resources Board regulations governing control of air pollution for new motor vehicles and new engines, in effect for this type of equipment on date of manufacture. The engine shall meet all operational requirements and conditions specified herein without exceeding the limits specfied in table I of MIL-STD-1410, or, the engine manufacturer's recommended operating limits. The average oil consumption of each engine shall not exceed .0035 pounds per brake-horsepower-hour (bhp/hr) when tested as specified herein.

3.13.1 Fuel system. A fuel tank shall be furnished. The capacity of the tank shall be sufficient for not less than 4 hours of continuous operation at the rated load. The fuel tank shall be equipped with a cap conforming to MS35645 and a strainer and neck conforming to MS35644. The tank shall be provided with a hand-operated fuel shutoff valve attached directly to the tank and a mechanically actuated fuel gage mounted on the tank. The fuel tank outlet shall be not less than 1/2 inch above the bottom of the tank. The fuel line shall be 5/16 inch OD and shall be provided with a 3-way selector valve and a 1/2-inch 20 SAE flared fitting for connection with an auxiliary fuel supply. A corrosion-resisting circular metal face-plate shall be mounted on the fuel selector valve and shall be marked "OFF", "UNIT TANK" and "AUXILIARY". The fuel tank shall be so located that when filling, no overflow will contact hot surfaces or any part of the electrical systems.

3.13.2 Cranking system. An electrical cranking system based on the use of a 12- or 24-volt electrical circuit energized by storage batteries shall be furnished.

3.13.3 Charging system. A rectified alternating-current-type battery-charging system shall be provided. Batteries shall conform to MS35000-1. The battery-charging alternator shall have a rating of not less than 35 amperes.

3.13.4 Control device system. The following engine shutoff-type control devices shall be provided and shall be set to energize at the engine manufacturer's recommended limits:

a. High-temperature control.
b. Low lubricating-oil pressure control.
c. Overspeed control.

3.13.5 Instrumentation. The engine shall be furnished with the following instruments mounted and connected in the instrument panel:

a. Gage, lubricating-oil pressure.
b. Indicator, temperature.
c. Tachometer - hour meter.
d. Indicator, battery-generator (ammeter or voltmeter).
e. Indicator, air cleaner restriction.

3.13.6 Engine housing. The engine shall be provided with metal housing that shall enclose the engine sides, top and end (except for openings for cooling

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air), and shall be fitted with ventilation panels. Housing shall be not less than 0.047 inch nominal thickness when made of steel and 0.064 inch nominal thickness when made of aluminum. Air deflectors, when utilized, shall be not less than 0.035 inch nominal thickness when made of steel and 0.051 inch nominal thickness when made of aluminum. Dry channels or other similar arrangements shall be provided to prevent the entry of rain or melted snow. Provisions shall be made for draining the oil pan and fuel tank to the outside of the housing.

3.13.6.1 Panels. The housing shall have removable or hinged side panels reinforced with stiffeners. Hinged panels shall be provided with latches to hold all panels in the open position. The latches shall require manual (not spring) action for closing. Removable panels shall be provided with quick-release fasteners. The panels shall provide access to all components of the engine. A hasp or hasps shall be provided on the panel(s) to enable padlocking of the housing.

3.13.6.2 Hinges and fasteners. Hinges shall be of the continuous (piano) type with brass or corrosion-resisting steel hinge pins. Hinges, latches, and quick-release fasteners shall be proportioned to facilitate manipulation by operators wearing arctic mittens.

3.14 Instrument panel. An instrument panel shall be located on the pump end of the engine housing. The panel shall have a hinged door and be flush with the housing. The panel shall contain all engine instruments specified herein, pump suction and discharge pressure gages and a battery-disconnect switch. The panel shall be provided with an interior switch-activated panel light.

3.14.1 Pressure gages. Two vibration-resistant pressure gages of non-oxidizing material conforming to ANSI B40.1 shall be provided. The discharge gage shall have a range from 0 to 100 pounds per square inch (psi) pressure. The compound suction gage shall have a range from 0 to 100 psi pressure and a range from 0 to 30 inches mercury vacuum. Each gage shall be accurate to within 2 percent of the total gage reading. The dials shall be white with black calibration lines and figures, and shall be not less than 4-1/2 inches in diameter. The gages shall be located on the instrument panel and shall be provided with a fluid-pressure dampener conforming to MIL-D-2940, type II, class 1, composition optional. Shutoff cocks shall be provided in gage lines. All gage lines shall be located and protected to prevent damage by personnel while handling during operation and service and shall have provisions for drainage to prevent freezing.

3.14.2 Battery-disconnect switch. The pump unit shall be provided with a battery-disconnect switch to permit disconnecting the battery in case of emergencies. The switch shall be installed in a convenient position on the instrument panel and shall withstand the full cranking motor load.

3.14.3 Performance data chart. A performance data chart approximately 8-1/2 inches by 11 inches shall be furnished. The chart shall show complete pump performance characteristics at the operating speed of the pump and performance

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of the pump at 25 feet static suction lift. All data shall be derived from the actual performance of the first article. The chart shall be of the same material used for identification plates. The chart shall be located on the inside of the instrument panel cover and shall be permanently secured in place by metal fasteners.

3.15 Electromagnetic interference. The electromagnetic interference emission characteristics of the pump shall conform to MIL-STD-461, class IIIC.

3.16 Lubrication. All surfaces requiring lubrication shall be provided with a means for lubricating.

3.16.1 Lubricants. The pump unit shall operate as specified herein when lubricated with military lubricants (see 6.4).

3.16.2 Lubrication fittings. Lubrication fittings shall conform to MS15003. Fittings shall be located in a protected position and shall be accessible to a grease gun conforming to MIL-G-3859. Accessibility to fittings shall be provided without the removal or adjustment of accessories or parts. Engine-housing side panels and plates equipped with hand-operable quick-disconnect fasteners may be removed to provide accessibility to lubrication fittings.

3.16.3 Pressure-release device. A pressure-release device shall be provided where the use of pressure lubricating equipment could damage grease seals or other parts.

3.16.4 Grease lubrication. All grease lubrication, including lubrication of sealed bearings, shall be with greases as follows:

a. MIL-G-23827 - For ball, roller, and needle bearings, gears, and on sliding and rolling surfaces of such equipment as instruments and mechanical control system; for use on gears, actuator screws and other equipment requiring a lubricant with high load-carrying capacity over a temperature range of -100° F to +250° F.
b. MIL-G-10924 - For lubrication and surface corrosion protection of all automotive and artillery equipment operated over the temperature range of -65° F to +225° F. Not for use on machinery which comes in contact with foods or potable water.

The pump shall be assembled, run-in (broken-in), tested, and delivered with these greases. The bearings shall be cleaned before lubrication because military greases are not always compatible with other greases. A tag shall be attached in a conspicuous place to indicate which military grease has been used.

3.16.5 Oil lubrication. Unless otherwise specified herein, all components shall be filled to the operating level with military oils designated for use in the temperature range down to 32° F. A tag shall be attached in a conspicuous place to indicate the temperature range and grade of lubricant used.

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3.17 Slinging provisions. Permanently affixed slinging provisions that enable the pump to be lifted in its normal travel position shall be provided and shall conform to MIL-STD-209. Slinging provisions may also be used as tiedowns when such provisions conform to MIL-STD-209.

3.18 Tiedown provisions. Permanently affixed tiedown provisions that permit fastening the pump to the floor or deck of a transportation medium shall be provided and shall conform to MIL-STD-209. Tiedown provisions may also be used as slinging provisions when such provisions conform to MIL-STD-209.

3.19 Identification marking. The pump unit shall be identified in accordance with MIL-STD-130. The marking shall be applied to the pump unit on a plate conforming to MIL-P-514, type I, style 1, composition C of type I, grade A, class 1 material. Plates shall be attached by screws, bolts or rivets in a location where the plate shall be both visible and legible.

3.20 Instruction plates. Each pump shall be equipped with instruction plates or diagrams, including warnings and cautions, describing any special or important procedures to be followed in assembling, operating, or servicing the pump. Instruction plates shall conform to MIL-P-514, type III, composition C, of type I, grade A, class 1 material. Plates shall be attached by screws, bolts or rivets in a location where the plate shall be both visible and legible.

3.20.1 Identification marking of pump inlet and outlet. The pump suction and discharge shall be identified by markings integrally cast in the pump or by plates specified in 3.20 permanently secured in place by means of metal fasteners.

3.21 Treatment and painting. The portions of the pump normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704, type A, color as specified (see 6.2).

3.22 Government-loaned property. A military 3/4-ton, 4-by-4 M37 series truck conforming to FED-STD-292 will be loaned by the Government (see 6.5).

3.23 Workmanship.

3.23.1 Castings and forgings. All parts, components, and assemblies of the pump unit which include castings and forgings shall be clean of harmful extraneous material such as sand, dirt, pits, sprues, scale, and flux. Rework shall be limited to procedures which do not reduce strength or affect function.

3.23.2 Metal fabrication. Metal used in fabrication shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the material. Corners shall be square and true. Flame-cutting, using tips suitable for the thickness of the steel may be employed instead of shearing and sawing. All bends shall be made with controlled means to insure uniformity of size and shape. Precaution shall be taken to avoid overheating. Heated steel shall be allowed to cool slowly. External surfaces shall be free of burrs, sharp edges and corners, except when sharp edges or corners are required or where they are not detrimental to safety.

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3.23.3 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, mill scale that can be removed by chipping and wire brushing, and other foreign matter. Parent materials, weld filler materials, and fabrication techniques shall be as required to enable the pump unit to conform to the examination and test requirements specified in section 4. Parts to be joined by fillet welds shall be brought into as close contact as possible and in no event shall be separated by more than 3/16 inch, unless appropriate bridging techniques are used. Unless otherwise specified (see 6.2), the welding process used in fabrication of the pump units shall be at the option of the contractor.

3.23.3.1 Welders. Before assigning any welder to manual welding work covered by this specification, the contractor shall provide the contracting officer with certification that the welder has passed qualification tests as prescribed by either of the following listed codes for the type of welding operations to be performed and that such qualification is effective as defined by the particular code:

AWS D1.1, Structural Welding Code, Section 5, Qualification.
ASME Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

Contractors who only make horizontal welds need not qualify welders for "all position welding." Subject to approval by the Government, the contractor's standard welder qualification may be substituted in lieu of the above codes provided that the contractor's procedure is equivalent to the above codes. The contractor shall be responsible for determining that automatic welder equipment operators are capable of producing quality welds.

3.23.4 Bolted connections. Boltholes shall be accurately formed and shall have the burrs removed. Washers or lockwashers shall be provided where necessary. Matching thread areas securing bolts conforming to SAE J429 or capscrews shall be of sufficient strength to withstand the tensile strength of the bolt. All fasteners shall be correctly torqued and shall have full thread engagement. Bolts shall not protrude more than two full threads.

3.23.5 Riveted connections. Rivets shall fill the holes completely. The upset rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member, and shall be in accordance with SAE J492.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications, standards, and drawings, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection comparison (see 4.6).
d. Inspection of packaging (see 4.7).

4.3 First article inspection. One or more first article pumps, as specified (see 6.2), will be selected by the Government from the pumps being produced by production tooling and will be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection. The first article pumps shall be tested as specified in column 1 of table I to determine conformance to the requirements of this specification. Failure of any test shall be cause for rejection.

4.3.1 Inspection failure. Failure of a first article pump to meet any requirement specified herein during and as a result of the examination and tests specified in 4.3, shall be cause for rejection of the first article pumps and shall be cause for refusal by the Government to continue acceptance of production pumps until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on pumps previously accepted and produced under the contract. Any deficiencies found as a result of the first article inspection will be considered prima facie evidence that all pumps accepted prior to the completion of first article inspection are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105, inspection level S-2.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined as specified in 4.5.1. AQL shall be 2.5 percent defective.

4.4.3 Tests.

4.4.3.1 Individual. Each pump unit shall be tested as specified in column 2 of table I.

4.4.3.2 Samples. Samples selected in accordance with 4.4.1 shall be tested as specified in column 3 of table I. AQL shall be 2.5 percent defective.

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4.5 Inspection procedure.

4.5.1 Examination. The pumps shall be examined as specified herein for the following defects:

101. Dimensions not as specified.
102. Materials not as specified or they are not resistant to corrosion and deterioration or treated to be made resistant to corrosion and deterioration for the applicable storage and operating environment.
103. Contractor does not have documentation available for identification of material, material finshes or treatments.
104. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.
105. All rotating parts and parts subject to high temperatures that are of such nature or so located as to become a hazard to operating personnel not insulated, fully enclosed, or guarded.
106. Fuel system not as specified.
107. Electrical system not as specified.
108. Cranking system not as specified.
109. Safety devices not as specified.
110. Instrument panel incorrectly located or not as specified.
111. Axle assembly and rated capacity of the axle not as specified.
112. Covers and plates not equipped with quick-disconnect fittings.
113. Drain outlets not furnished or not as specified.
114. Pump-case drain port and filling provision not as specified.
115. Pump not equipped with corrosion-resistant, grease-lubricated, mechanical seal.
116. Intermediate bearing housing not as specified.
117. Pump case not as specified.
118. Pump not provided with wear plate.
119. Pump impeller not as specified.
120. Engine controls not provided or not as specified.
121. Engine instruments not provided or not as specified.
122. Weights and loads not as specified.
123. Lunette and support legs not furnished or not as specified.
124. Coupling not as specified.
125. Lubrication fittings not as specified or not accessible.
126. Toolbox not provided or not as specified.
127. Raising, lowering, and locking action of support legs not as specified.
128. Safety chains, intervehicluar hoses, and cables not as specified.
129. Reflectors improperly located, missing, or defective.
130. Chock blocks and cradles not furnished or not as specified.
131. Lifing or tiedown provisions not furnished.
132. Fuel tank not as specified.
133. Screw threads not as specified.
134. Workmanship not as specified.
135. Identification marking not as specified.
136. Instruction plates missing or not as specified.
137. Improper coding, protection, or installation of wiring and tubing.
138. Painting not as specified.
139. Lubricants not as specified.
140. Electromagnetic interference suppression components and network not identical to the approved model.

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4.5.2 Tests.

4.5.2.1 Test conditions. Prior to testing, the pumping unit shall be lubricated with oils and greases specified herein. Oils shall be those designated for use in the ambient temperature at the place of test. Tests shall be performed in accordance with the applicable provisions of the test code for centrifugal pumps of the Standards of the Hydraulic Institute. Unless otherwise specified herein, the tests shall be conducted using a 10-foot static suction lift. Water shall be used as the test liquid, and all test data shall be corrected to standard sea level atmospheric conditions of 29.92 inches of mercury and to the specific weight of water (62.3 pounds per cubic foot) at a temperature of 68° F. The hydrostatic and the head capacity tests on the first article shall be performed prior to, and after completion of, the mobility characteristics test.

4.5.2.1.1 Test schedule.

TABLE I. Test schedule.

First Article	Quality Conformance		Tests	Test Paragraph	Requirement Paragrpah
	Individual	Sample			
1	2	3	4	5	6
X	X	-	Pump impeller balance.	4.5.2.2.1	3.10.3
X	-	-	Pump BHP.	4.5.2.2.2	3.13.1
X	X	-	Hydrostatic.	4.5.2.3.1	3.10.7
X	X	-	Priming.	4.5.2.3.2	3.10.8.1
X	X	-	Head capacity.	4.5.2.3.3	3.10.8.2
X	-	X	Check valve.	4.5.2.3.4	3.10.6
X	-	-	Operational 100 hour.	4.5.2.3.5.1	3.10.8
-	-	X	Production.	4.5.2.3.5.2	3.10.8
X	-	-	Mobility.	4.5.2.3.6	3.11.1, 3.11.2
X	-	-	Slinging.	4.5.2.3.7	3.17
X	-	-	Tiedown.	4.5.2.3.8	3.18
X	-	-	Electromagnetic.	4.5.2.4	3.15
X	-	-	Noise level.	4.5.2.5	3.7.1
X	-	-	Reliability.	4.5.2.6	3.9
X	-	-	Maint. spt. eval.	4.5.2.7.1	3.8.1
X	-	-	Maint. ratio	4.5.2.7.2	3.8.2
X	-	-	Safety.	4.5.2.8	3.6
X	-	-	High temperature.	4.5.2.9	3.13

4.5.2.2 Preassembly test.

4.5.2.2.1 Pump impeller balance. During assembly of the pump unit, determine the "out of balance" of the impeller. Out of balance of more than 0.25 oz-in shall constitute failure of this test.

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4.5.2.2.2 Pump brake horsepower determination. Prior to assembly of the pump to the engine, operate the pump, including bearing housing, as specified in 3.10.8.2, using a dynamometer or calibrated electric motor to determine the brake horsepower required by the pump. A required brake horsepower in excess of the continuous horsepower rating of the engine (see 3.13.1), or a peak pump efficiency of less than 65 percent shall constitute failure of this test.

4.5.2.3 Assembled tests.

4.5.2.3.1 Hydrostatic. Subject the pump to a hydrostatic pressure equal to 1-1/2 times the shutoff head obtained at full throttle operation. Leakage or permanent deformation of the pump shall constitute failure of this test.

4.5.2.3.2 Priming. Prime and operate the pump unit in accordance with conditions specified in 3.10.8.1, except the first article model shall be tested for both the 10- and 25-foot static suction lift. Nonconformance to 3.10.8.1 shall constitute failure of this test.

4.5.2.3.3 Head capacity. Prime and operate the pump to determine conformance to the head capacities shown in figure 1 not exceeding 2600 rpm. Nonconformance to 3.10.8.2 shall constitute failure of this test.

4.5.2.3.4 Check valve. After the pump has been completely primed, subject the check valve to a maximum test pressure (total static discharge head) equal to the maximum total head as shown in figure 1. Shut off the pump and allow it to set for not less than 30 minutes. Restart the pump. Determine that the pump primes itself. Self-priming of the pump is an indication that the valve prevents siphoning of water from the pump case. Examine the check valve. Inability of the pump to prime itself, leakage, or permanent deformation of any component of the valve shall constitute failure of this test.

4.5.2.3.5 Operational.

4.5.2.3.5.1 100-hour. Prime the pump in accordance with 4.5.2.3.2 using a static suction lift equivalent to 10 feet. Operate the pump at the head-capacity condition requiring the maximum brake horsepower determined during test under 4.5.2.2.2 for 100 hours continuously. During operation, only minor adjustments to the engine and discharge valving may be performed. A maximum shutdown time of 15 minutes is allowable for normal servicing, maintenance and examination during each 12-hour period. During operation, determinations shall be made as to whether there is leakage through the pump case, pump flange, piping, fittings, gages, or at the pump shaft seal, and whether there is malfunction of any component. Midway through the test determine whether all the safety devices are functioning. Evidence of leakage, malfunction of any component, inability of the pump to prime in less than 1 minute or to deliver the capacity established at the beginning of the test, inability of the safety devices to perfrom the function intended, or inability of the pump unit to operate continuously (except for authorized shutdown) shall constitute failure of this test.

4.5.2.3.5.2 Production. Test the pump unit as specified in 4.5.2.3.5.1, except that the operating time shall be 30 minutes and shall be at the flow rate of 1500 gpm.

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4.5.2.3.6 Mobility characteristics. Couple the pump unit to the Government-loaned truck specified in 3.22. Tow the pump unit over roads and the speeds specified in 3.11.1 for the following distances:

a. Cross-country operation - 50 miles.
b. Unimproved-road operation - 150 miles.
c. Paved-highway operation - 200 miles.

During the towing, observe the tracking of the pump unit. Make 25 right-circle turns and 25 left-circle turns, all with the steering of the towing vehicle turned to lock position. Determine the turning clearance. Examine the pump unit at completion of this test. Nonconformance to 3.11.1 and 3.11.2 shall constitute failure of this test.

4.5.2.3.7 Slinging provisions.

a. Lift the pump and hold it in suspension in its normal travel position using slings that converge not more than 24 feet above the lowest extremity of the pump. Determine the forward angles of application for each provision. Measure the clearance between each sling and the pump. Measure the eye openings and clearance dimensions of the provisions.
b. Restrain the pump by anchoring the main frame and subject each slinging provision to a force of 2-1/2 times the load it carried when suspended. This force shall be applied in the direction as determined in "a" above. An acceptable alternate method of test shall be to lift the pump as in "a" above and add weights to the pump until the force on each provision is 2-1/2 times the initial force. Hold the pump under either of the above loading conditions for not less than 90 seconds.

Weld failure, permanent deformation of the pump or any slinging provision, slings converging more than 24 feet above the lowest extremity of the pump, or nonconformance to MIL-STD-209 shall constitute failure of this test.

4.5.2.3.8 Tiedown provisions. With the pump anchored by means other than the tiedown being tested, apply the static loads specified in MIL-STD-209 to each provision. Measure the dimensions of the tiedowns. Weld failure, permanent deformation of the pump or its tiedown provisions, breakage of the provisions, or nonconformance to MIL-STD-209 shall constitute failure of this test.

4.5.2.4 Electromagnetic interference. The first article pump unit, equipped for the reduction of radio interference in accordance with 3.15, shall be tested in accordance with MIL-STD-462, test method RE05. The contractor shall furnish the contracting officer the test reports required in MIL-STD-461. Disapproval of the reports shall constitute failure of this test.

4.5.2.5 Noise level measurement. Measure the noise level of the pump in accordance with MIL-STD-1474. Nonconformance with 3.7.1 shall constitute failure of the test.

4.5.2.5.1 Hazard precautions. Measure the noise level at the control position while operating at the maximum noise condition at that location.

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Should the noise level exceed 85 dB(A), failure to provide the precautions specified in 3.7.1 shall constitute failure of this test.

4.5.2.6 Reliability. The pump shall be operated in accordance with the conditions specified in 4.5.2.1 at not more than 2600 rpm and a flow rate of 1500 gpm. Operations and maintenance shall be performed in accordance with the instruction set forth in the manufacturer's operation and maintenance schedule. Using the MTBF specified in 3.9, the pump shall be tested as specified with "accept" and "reject" criteria in accordance with figure 2. Continue testing until either an "accept" or "reject" decision is reached. A failure is defined as any malfunction which cannot be corrected with 30 minutes (all waiting periods are excluded by adjustment, repair, or replacement action using controls and on-equipment tools or parts available to the using element and is a malfunction which causes or may cause one or more of the following:

a. Failure to commence operation, cessation of operation, or degradation of performance below the designated levels.
b. Serious damage to the system.
c. Serious personnel safety hazards.

Nonconformance to 3.9 shall constitute failure of this test.

4.5.2.7 Maintainability.

4.5.2.7.1 Maintenance support evaluation. Maintenance operations required during first article testing shall be accomplished to determine conformance to 3.8.1. Failure of the support elements to maintain the pump as specified shall constitute failure of this test. Errors or inadequacies in the manuals shall not be considered in assessing the maintenance ratio or the MTBF.

4.5.2.7.2 Maintenance ratio. The maintenance ratio shall be computed during testing specified in 4.5.2.6 and 4.5.2.9. Nonconformance to 3.8.2 shall constitute failure of this test.

4.5.2.8 Safety. The pump shall be evaluated for safety throughout the testing specified in 4.5.2.3.6 and 4.5.2.6. Nonconformance to 3.6 shall constitute failure of this test.

4.5.2.9 High temperature test. Set up the pump near a water source of sufficient size to supply water so that the pump can perform at its maximum load condition without overheating the pumped water during a 2-hour operating period. Completely enclose the entire pump assembly in a shelter or container so that the heat from the engine coolant will raise the temperature inside the enclosure to 120° F. Sufficient vents shall be provided in the enclosure so that the temperature may be stabilized. Engine exhaust shall be conveyed outside. After the temperature has stabilized at not less than 120° F, operate the pump at its maximum load condition as shown on its performance curve for 2 hours. Nonconformance to any requirement in 3.13 or any loss in performance or malfunction of the engine or pump shall constitute failure of this test.

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4.6 Inspection comparison. The Government may select pumps at any time during the contract production period and subject these pumps to the examination and the tests specified in 4.3 to determine conformance to the requirments of this specification. The inspection will be performed by the Government, at a site selected by the Government, on pumps selected at random from those which have been accepted by the Government and will not include the previously inspected first article model pumps. In addition to any test specified as part of the inspection comparison, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison, and failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

4.6.1 Inspection failure. Failure of an inspection comparison pump to meet any requirement specified herein during and as a result of the examination and tests specified in 4.6 shall be cause for rejection of the inspection comparison pump and shall be cause for refusal by the Government to continue acceptance of production pumps until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on pumps previously accepted and produced under the contract. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all pumps accepted prior to the completion of inspection comparison are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.7 Inspection of packaging. The preservation, packing, and marking shall be examined to determine compliance with the quality assurance provisions of MIL-P-10603.

5. PACKAGING

5.1 Preservation, packing and marking. Each complete pump and all associated components shall be preserved, packed and marked in accordance with MIL-P-10603 except reference therein to FED-STD-356 shall be corrected to reference MIL-STD-1188. The degree of preservation and packing shall be as specified (see 6.2).

6. NOTES

6.1 Intended use. The pump units are intended for general purpose water pumping applications associated with construction work.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Time frame required for submission of first article model and number of pumps to be furnished (see 3.2).
c. When the Government will conduct any or all of the first article model examination and tests. When the Government will conduct some but not

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all of the first article examination and tests, the contracting officer should specify which examination and tests will be conducted by the Government and which examination and tests shall be conducted by the contractor (see 3.2).

d. Color required (see 3.21).
e. When the welding process shall not be at the option of the contractor (see 3.23.3).
f. Degree of preservation and degree of packing required (see 5.1).

6.3 First article model. Any changes or deviations of production pumps from the approved first article model during production will be subject to the approval of the contracting officer. Approval of the first article model will not relieve the contractor of his obligation to furnish pumps conforming to this specification.

6.4 Lubricants. The contracting officer should furnish a list of military lubricants applicable to the pump covered by this specification as contained in the Federal Supply Catalogue, Department of Defense Section, Identification List C9100-IL, for FSC Group 91.

6.5 Government-loaned property. The contracting officer should arrange to loan the property specified in 3.22.

6.6 Definitions. The following definitions shall apply throughout this specification.

6.6.1 Cross-country operation. Cross-country operation shall consist of traveling over open fields, broken ground, loose sand or gravel, and uneven terrain in general.

6.6.2 Unimproved road operation. Unimproved road operation shall consist of traveling over level or rolling gravel roads.

6.6.3 Highway operation. Highway operation shall consist of traveling over level or rolling paved roads.

6.6.4 Specified mean-time-between-failure (MTBF O_o). Specified mean-time-between-failure is the MTBF value specified in the contract or equipment specification.

6.6.5 Producers decision risk (a). Producers decision risk is the probability of rejecting equipment(s) with a true MTBF equal to the specified MTBF (O_o).

6.6.6 Comsumers decision risk (B). Consumers decision risk is the probability of accpeting equipment(s) with a true MTBF equal to the minimum acceptable MTBF (O_1).

6.6.7 Discrimination ratio (O_o/O_1). Discrimination ratio is the ratio of specified MTBF (O_o) to the minimum acceptable MTBF (O_1).

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6.7 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and contractor's maintenance and operation manual to be furnished with each pump.

6.8 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools), and instructions regarding shipment of pumps.

6.9 Recycled material. It is encourage that recycled material be used when practical as long as it meets the requirements of the specification (see 3.3).

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CX-1095

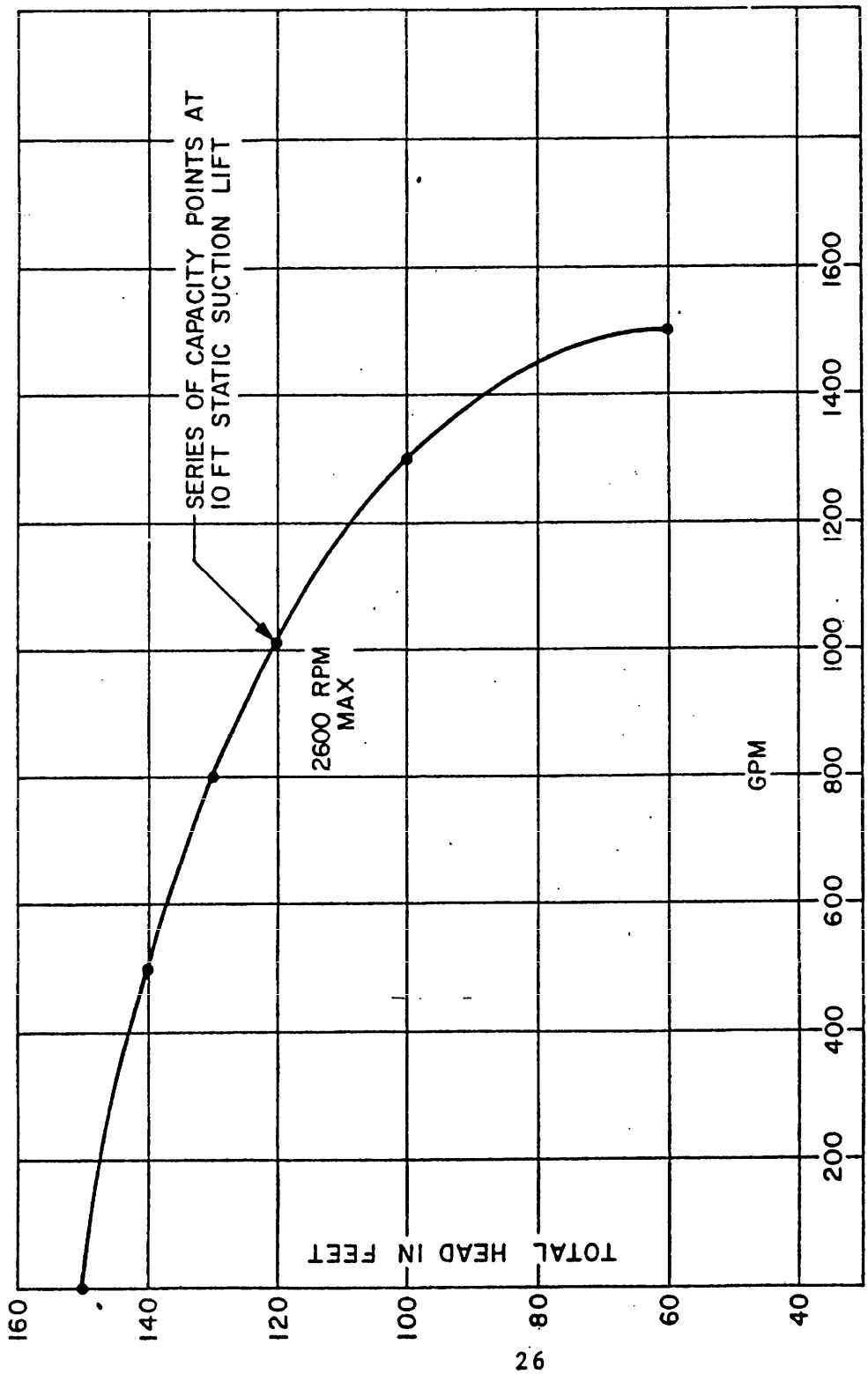
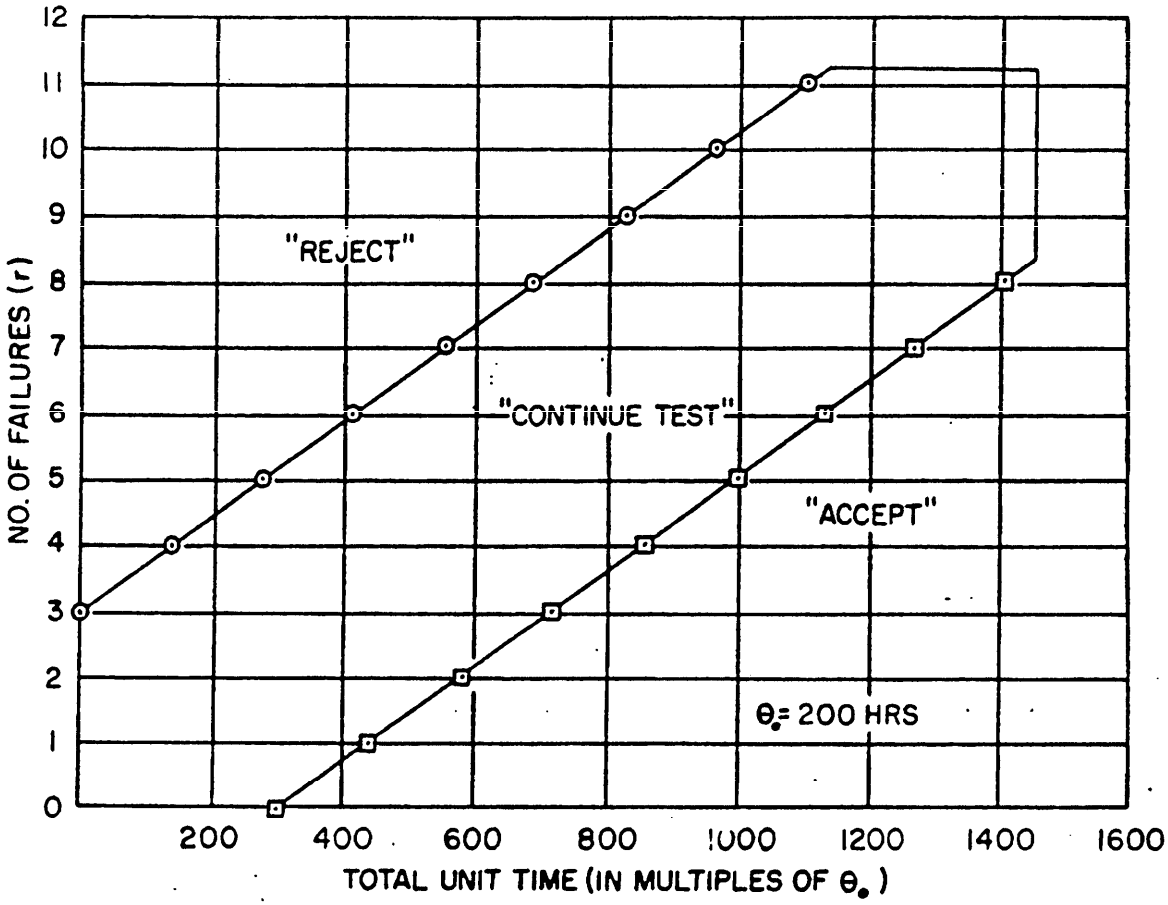


FIGURE 1. PUMP CHARACTERISTICS

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a - PRODUCERS DECISION RISK - 10%
b - CONSUMERS DECISION RISK - 20%
c - DISCRIMINATION RATIO θ_o/θ_1 - 2.00

TABLE III
ACCEPTANCE TESTING CRITERIA

r	REJECT (EQUAL OR LESS)	ACCEPT (EQUAL OR MORE)
0	N/A	302
1	N/A	440
2	N/A	578
3	0	716
4	138	856
5	278	994
6	416	1132
7	544	1270
8	692	1410
9	832	1460
10	970	1460
11	1108	1460
12	1460	N/A

FIGURE 2. Accept-reject criteria for reliability testing.

X-2327A

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-52565D(ME)	Pump, Centrifugal: Self-Priming, Diesel-Engine-Driven, Wheel Mounted, 6-Inch, 1500 GPM at 60 Foot Head

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional

b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional

c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional

8. DATE OF SUBMISSION *(YYMMDD)*

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